

# Math Basics

Session 1: (Statistics Fundamentals)

## Agenda:

|   |                                                  |
|---|--------------------------------------------------|
| 1 | Introduction to Statistics (30min)               |
| 2 | Descriptive Statistics (1 hour)                  |
| 3 | Probability Basics (1 hour)                      |
| 4 | Introduction to Inferential Statistics (30 mins) |

# Introduction to Statistics (30min)

## Importance of Statistics in AI and Machine Learning

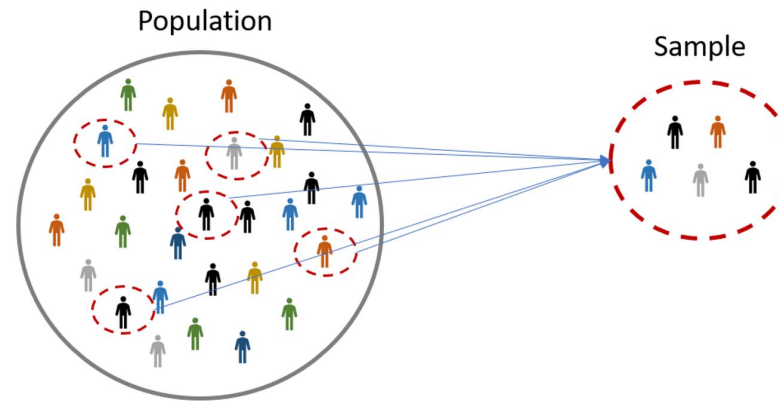
### Why Learn Statistics for AI?

- **Data Analysis:** Statistics provide the tools to analyze and interpret data, which is the backbone of AI. Understanding data distributions, variability, and patterns is crucial for building robust models.
- **Model Evaluation:** In AI, especially in machine learning, evaluating the performance of models (e.g., accuracy, precision, recall) relies on statistical measures. Without a strong understanding of statistics, interpreting these metrics can be challenging.
- **Decision Making:** AI models often make decisions based on probabilities. For example, a classification model might predict the likelihood of an email being spam. Understanding probability and statistical inference is key to trusting and explaining these decisions.
- **Real-World Applications:** From predicting customer behavior in marketing to diagnosing diseases in healthcare, statistics provide the foundation for analyzing data and making informed decisions.

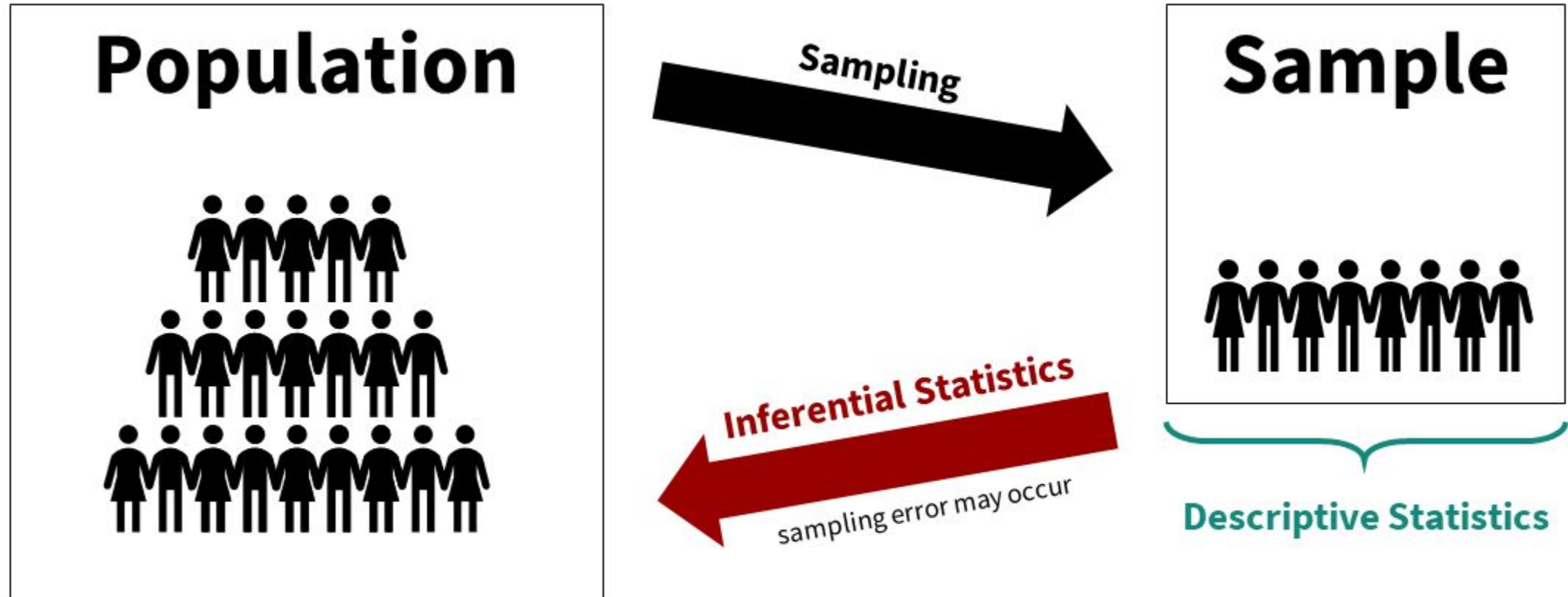
## Basic Statistical Definitions

### Population vs. Sample

- **Population:** The entire group of individuals or items that you're interested in studying. For example, if you want to study the average height of adult males in a country, the population would be **all** adult males in that country.
- **Sample:** A subset of the population that is used to represent the whole. For instance, measuring the height of **1,000 adult males** from different regions could be a sample representing the entire population. Sampling is often done because it is impractical or impossible to study the whole population.



# Descriptive statistics vs Inferential statistics



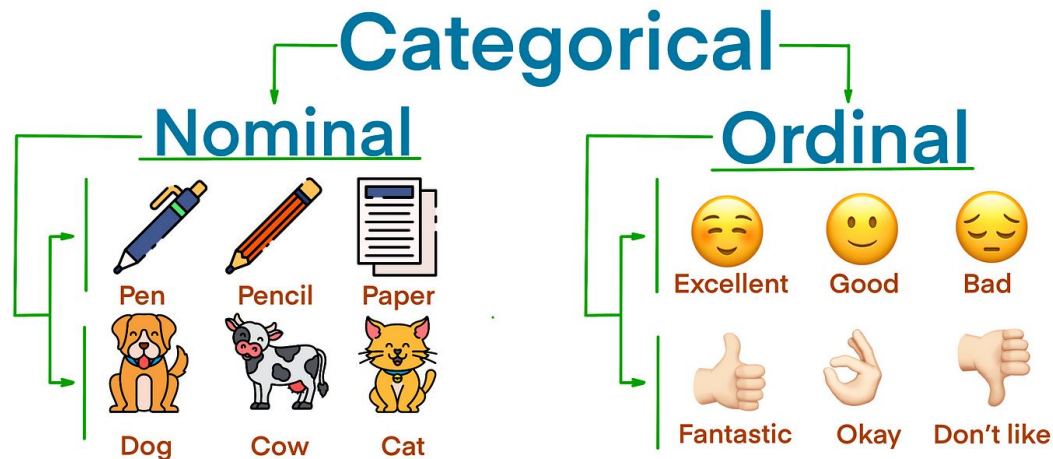


## Types of Variables

### Categorical (Qualitative) Variables

These variables represent categories or groups and can be either:

- **Nominal:** No inherent order among categories. Examples include colors (red, blue, green), gender (male, female).
- **Ordinal:** There is an inherent order, but the difference between levels is not quantifiable. Examples include customer satisfaction ratings (poor, fair, good, excellent), education levels (high school, bachelor's, master's).



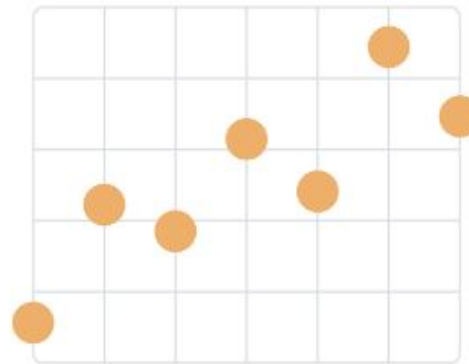
## Types of Variables

### Numerical (Quantitative) Variables

These variables represent measurable quantities and can be either:

- **Discrete:** Countable, finite values. Examples include the number of students in a class, the number of cars in a parking lot.
- **Continuous:** Infinite possibilities within a range. Examples include height, weight, temperature.

DISCRETE DATA



CONTINUOUS DATA





## Real-World Relevance and Applications

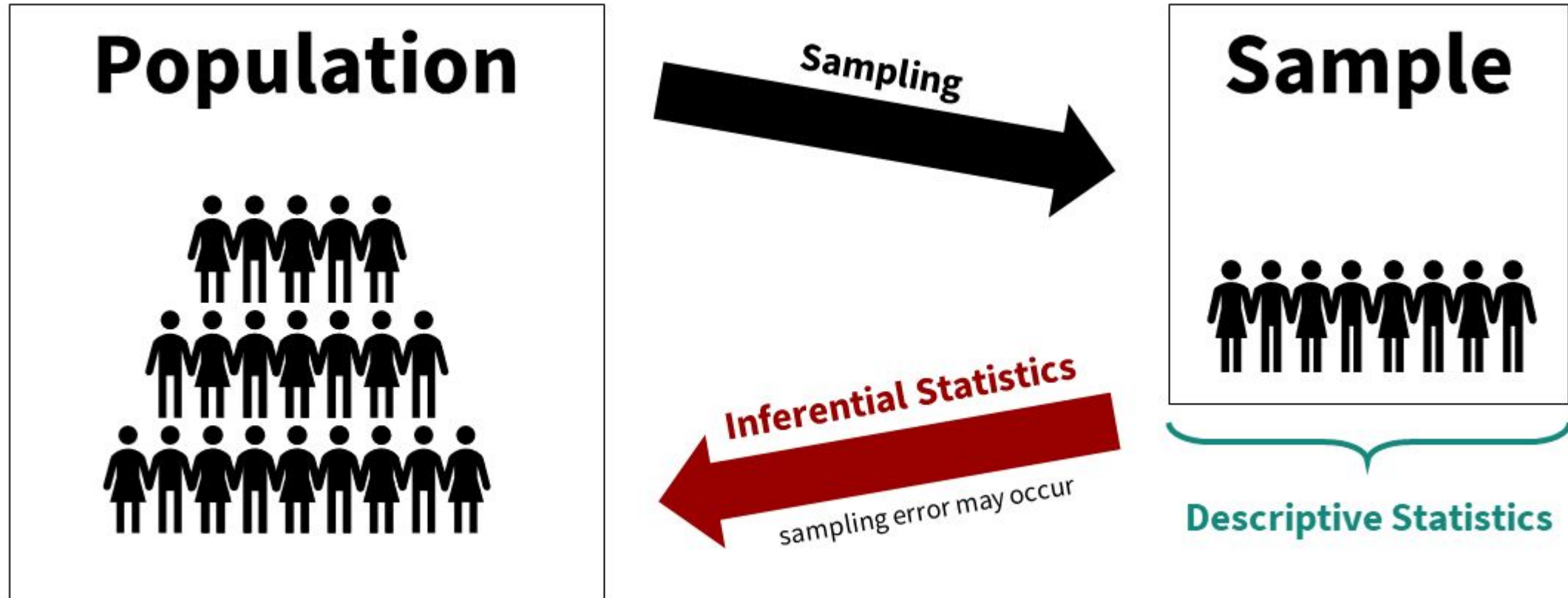
### Examples of Statistics in AI Applications

- **Fraud Detection:** Analyzing transaction data to identify patterns that indicate fraudulent activity. Here, statistics help to detect anomalies that deviate from normal behavior.
- **Recommendation Systems:** E-commerce platforms like Amazon use statistics to analyze customer preferences and recommend products based on what similar customers have purchased.
- **Healthcare Analytics:** Predicting the likelihood of diseases based on patient data. Statistical models can assess risk factors and provide personalized healthcare recommendations.



# Descriptive Statistics (1 hour)

# Descriptive statistics vs Inferential statistics



## Descriptive Statistics

Descriptive statistics help provide a clear understanding of data through numerical calculations, graphs, and tables. This is a crucial step before conducting any further statistical analysis or building machine learning models.

## Cases vs. Variables

### Cases

- A case represents an individual entity or subject in a dataset on which measurements or observations are made. It can be a person, a country, an event, or any other subject of study.

### Variables

- A variable is a characteristic or attribute that can be measured or observed for each case. Variables are features or properties that describe the cases

|       |   | Variables |       |           |         |
|-------|---|-----------|-------|-----------|---------|
|       |   | Store     | Sales | Customers | Refunds |
| Cases | 1 | 14        | 9     | 3         |         |
|       | 2 | 19        | 13    | 4         |         |
|       | 3 | 22        | 19    | 4         |         |
|       | 4 | 24        | 20    | 3         |         |
|       | 5 | 29        | 26    | 8         |         |
|       | 6 | 40        | 34    | 6         |         |

## Data Matrix vs. Frequency Table

### Data Matrix

A data matrix is a structured table where each row represents a single case (individual data point), and each column represents a variable (characteristic or feature of the cases).

| Variables    |             |              |             |                 |
|--------------|-------------|--------------|-------------|-----------------|
| sepal length | sepal width | petal length | petal width | class           |
| 5.1          | 3.5         | 1.4          | 0.2         | Iris-setosa     |
| 4.9          | 3           | 1.4          | 0.2         | Iris-setosa     |
| 6.5          | 3.2         | 5.1          | 2           | Iris-virginica  |
| 6.4          | 2.7         | 5.3          | 1.9         | Iris-virginica  |
| 6.8          | 3           | 5.5          | 2.1         | Iris-virginica  |
| 6.7          | 3.1         | 4.4          | 1.4         | Iris-versicolor |
| 5.6          | 3           | 4.5          | 1.5         | Iris-versicolor |
| 5.8          | 2.7         | 4.1          | 1           | Iris-versicolor |



## Data Matrix vs. Frequency Table

### Data Matrix

#### Purpose

The data matrix is useful for raw data representation where each case's specific details are important. It's a comprehensive way to store all the information but not necessarily the best for summarizing data.

#### When to Use

Data matrices are used in scenarios where you need to keep all details about each individual case, such as during data collection or when you need to perform case-by-case analysis.

## Data Matrix vs. Frequency Table

### Frequency Table

A frequency table is a summary of the data that shows how often each value of a variable occurs. It can display frequencies, percentages, and cumulative percentages.

| Score | Frequency |
|-------|-----------|
| 50-59 | 2         |
| 60-69 | 2         |
| 70-79 | 6         |
| 80-89 | 7         |
| 90-99 | 3         |

## Data Matrix vs. Frequency Table

### Frequency Table

#### Purpose

Frequency tables are used to summarize and visualize the distribution of a variable, making it easier to see patterns and understand the data's structure.

#### When to Use

When you want to quickly grasp how data is distributed across different categories or ranges, especially when dealing with categorical or quantitative variables.

## Descriptive Statistics

### 2 Types of Descriptive Statistics



#### Measures of central tendency

- Mean
- Median
- Mode

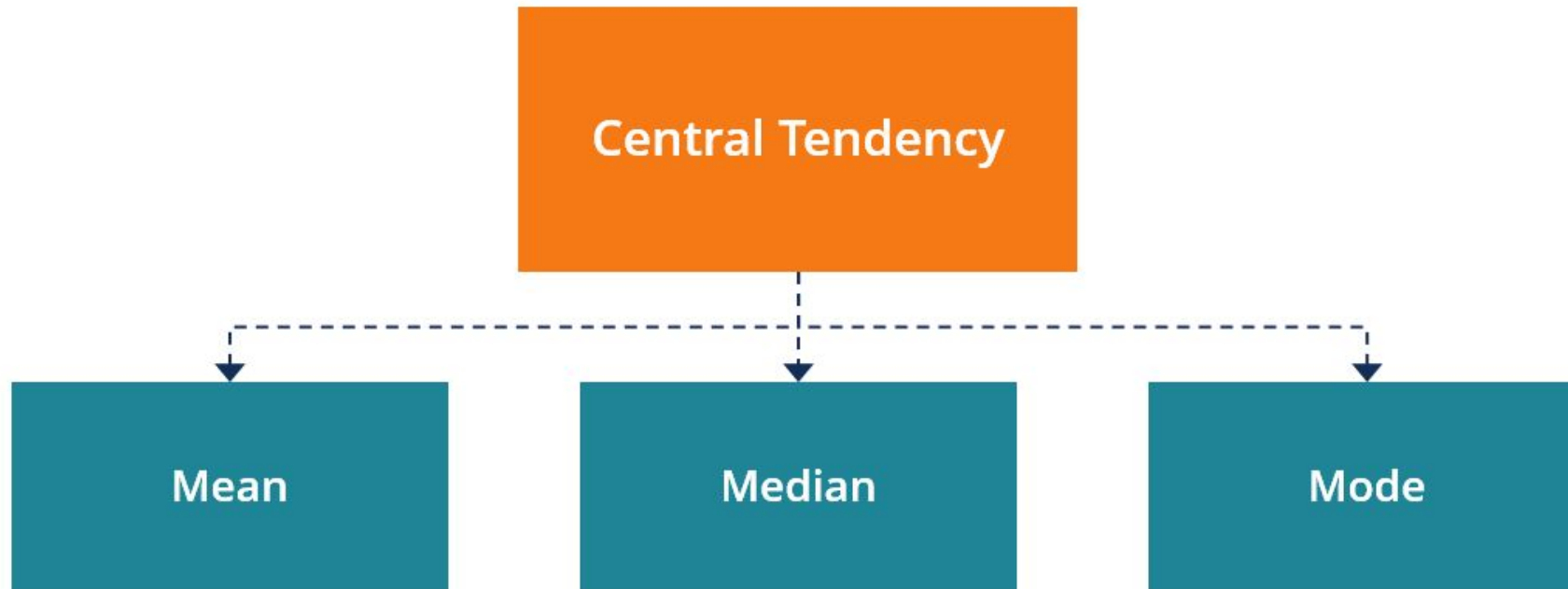
#### Measures of dispersion

- Range
- Interquartile range
- Variance
- Standard Deviation

## Descriptive Statistics

### Measures of Central Tendency

Understand how to describe the center of a dataset using different measures and learn how to calculate and interpret these values.



## Measures of Central Tendency

### Mean

#### (Average)

The mean is the sum of all values divided by the number of values. It is a commonly used measure of central tendency.

### Formul

a

For a dataset with values  $x_1, x_2, \dots, x_n$ , the mean  $\mu$  is calculated as

$$\mu = \frac{\sum_{i=1}^n x_i}{n}$$

### Example

For the dataset  $[4, 8, 6, 5, 3, 4]$ , the mean is

$$\mu = \frac{4 + 8 + 6 + 5 + 3 + 4}{6} = 5$$

The mean is sensitive to outliers. If the dataset has extreme values (very high or very low), the mean can be misleading



## Measures of Central Tendency

### Median

The median is the middle value of a dataset when it is ordered from smallest to largest. If the dataset has an even number of values, the median is the average of the two middle values.

### Example

For the dataset [4,8,6,5,3,4], first sort the data:[3,4,4,5,6,8]. The median is the average of the two middle numbers, 4 and 5, which  $\frac{4+5}{2} = 4.5$

The median is not affected by **outliers** and provides a **better measure** of central tendency for **skewed distributions**.

## Measures of Central Tendency

### Mode

The mode is the value that appears most frequently in a dataset.

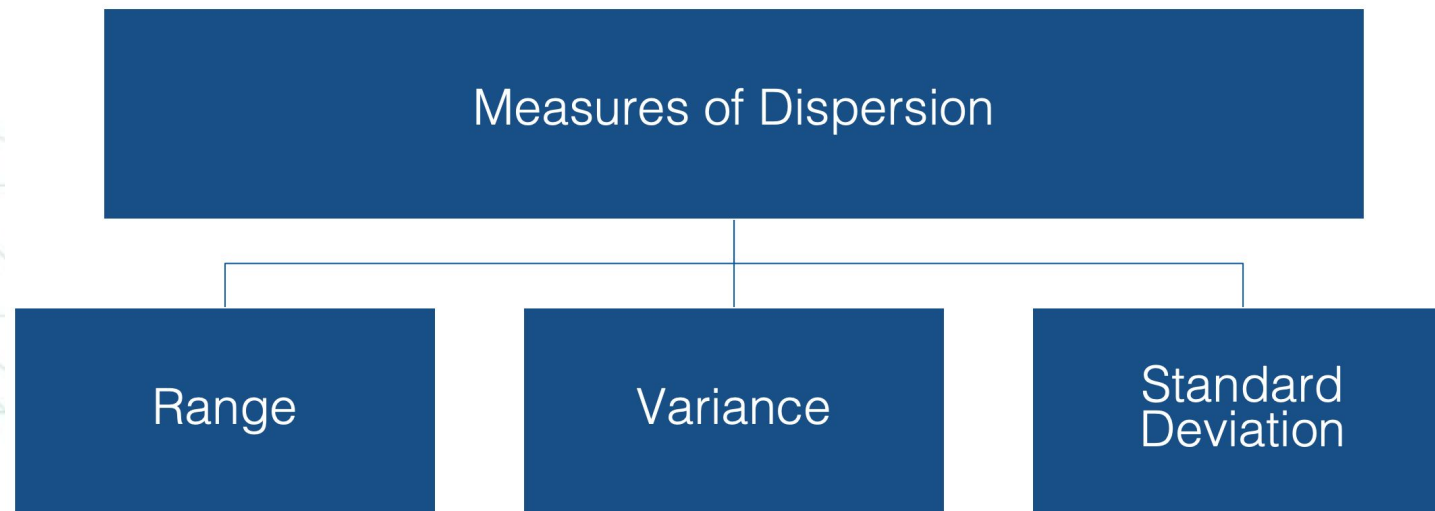
### Example

In the dataset  $[4, 8, 6, 5, 3, 4]$ , the mode is 4 because it appears twice, more than any other number.

The mode is useful for **categorical data** and for identifying the most common item in a dataset. A dataset can have more than one mode (bimodal or multimodal) or no mode at all.

## Measures of Dispersion

Learn how to describe the **spread** or **variability** of a dataset using different measures.



## Measures of Dispersion

### Range

The range is the difference between the **maximum** and **minimum** values in a dataset.

### Formula

Range=Max value–Min value.

### Example

For the dataset [4,8,6,5,3,4], the range is  $8-3=5$ .

The range gives a quick sense of the spread of the data but is **highly** sensitive to outliers.

## Measures of Dispersion

### Interquartile Range (IQR)

The IQR is the range between the first quartile (Q1) and the third quartile (Q3). It measures the spread of the middle 50% of the data, effectively capturing the central tendency without being influenced by extreme values or outliers. The IQR is calculated as

$$IQR = Q3 - Q1$$

- **Q1 (First Quartile):** The value below which 25% of the data falls.
- **Q3 (Third Quartile):** The value below which 75% of the data falls.

## Interquartile Range (IQR)

### Why Use IQR?

- **Robustness Against Outliers:** Unlike the range, which considers only the minimum and maximum values, the IQR focuses on the middle 50% of the data. This makes it less sensitive to outliers and extreme values, providing a more reliable measure of dispersion for skewed distributions.
- **Understanding Data Spread:** The IQR helps understand the variability of the central portion of the data. A larger IQR indicates more spread in the middle 50% of the dataset, while a smaller IQR suggests that the data points are closer to the median.



## Interquartile Range (IQR)

### Example Calculation of IQR

Let's use a small dataset to illustrate:

**Dataset:** [2,4,4,5,6,8,9]

1. **Arrange the data in ascending order** (already sorted in this case).
2. **Find Q1 and Q3:**
  - **Q1 (First Quartile):** The median of the first half of the dataset (excluding the median if the number of data points is odd). For [2,4,4],  $Q1 = 4$
  - **Q3 (Third Quartile):** The median of the second half of the dataset. For [6,8,9],  $Q3 = 8$

3. **Calculate IQR:** 
$$IQR = Q3 - Q1 = 8 - 4 = 4$$

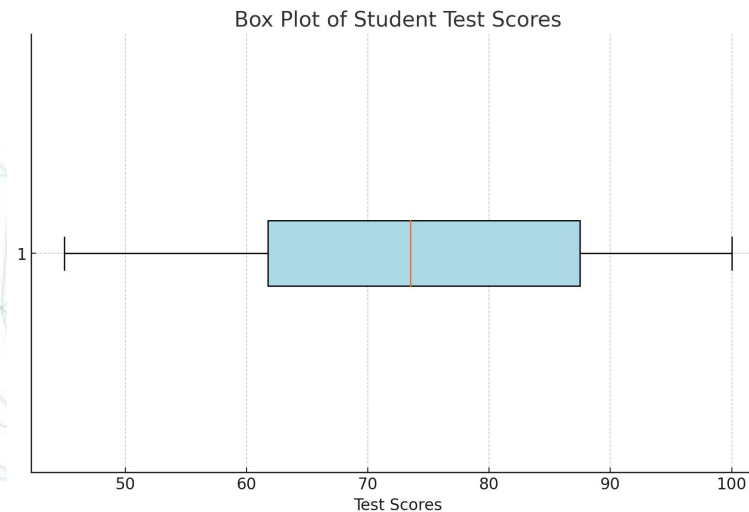
## Data Distribution and Visualization

### Box Plots (Box-and-Whisker Plots)

A box plot displays the distribution of data based on a five-number summary: minimum, first quartile (Q1), median (Q2), third quartile (Q3), and maximum.

#### Example

Create a box plot for a dataset of test scores to identify the median score, quartiles, and potential outliers.



- Box plots are useful for detecting outliers and understanding the spread and symmetry of the data.

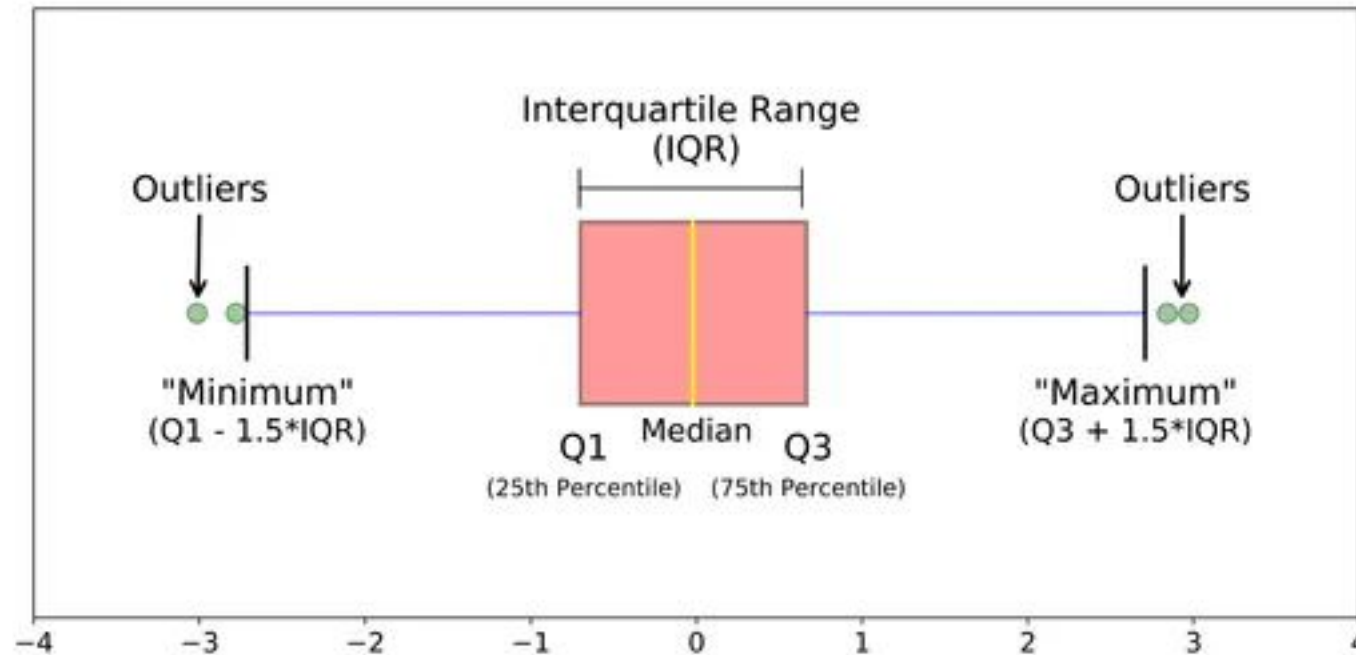
#### Quartiles

**First Quartile (Q1, 25th percentile):** The left edge of the box represents the 25th percentile, indicating that 25% of the students scored below this value.

**Third Quartile (Q3, 75th percentile):** The right edge of the box represents the 75th percentile, showing that 75% of the students scored below this value.

# Box Plot

When reviewing a box plot, an outlier is defined as a data point that is located outside the whiskers of the box plot.



## Measures of Dispersion

### Variance

Variance measures the **average squared deviation** of each number from the **mean**. It provides insight into the spread of all data points around the mean.

### Formula

For a dataset with values  $x_1, x_2, \dots, x_n$ , the variance  $\sigma^2$  is calculated as 
$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n}$$

### Example

For the dataset [4,8,6,5,3,4], calculate the mean ( $\mu=5$ ), then compute the variance:

$$\sigma^2 = \frac{(4-5)^2 + (8-5)^2 + (6-5)^2 + (5-5)^2 + (3-5)^2 + (4-5)^2}{6} = \frac{1+9+1+0+4+1}{6} = 2.67$$

Variance is in squared units, making it less interpretable in the original scale. However, it is fundamental in statistical theory.

## Measures of Dispersion

### Standard Deviation

The standard deviation is the **square root** of the **variance** and provides a measure of the **average distance** from the mean. It is in the same units as the data, making it more **interpretable**.

Formula:  $\sigma = \sqrt{\sigma^2}$

### Example

For the dataset [4,8,6,5,3,4], the standard deviation is  $\sigma = \sqrt{2.67} \approx 1.63$

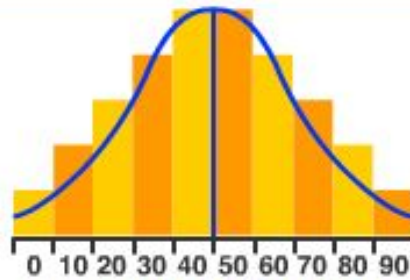
A small standard deviation indicates that the values are close to the mean, while a large standard deviation indicates that the values are spread out over a wider range.



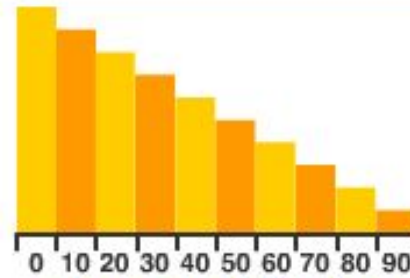


## Descriptive Statistics

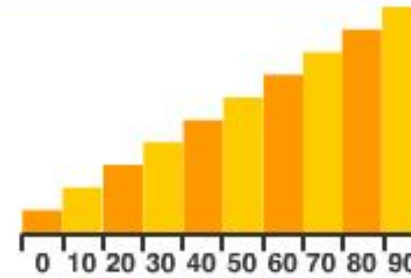
### Data Distribution and Visualization



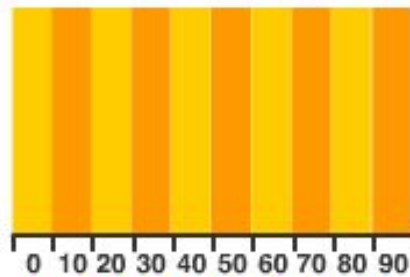
**normal distribution**  
unimodal, symmetric,  
aka 'bell curve'



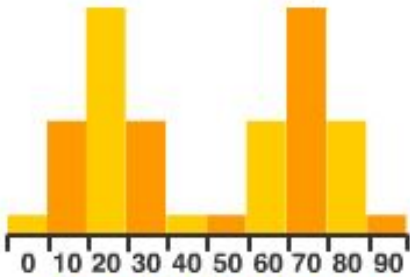
**skewed distribution**  
positively skewed,  
skewed right



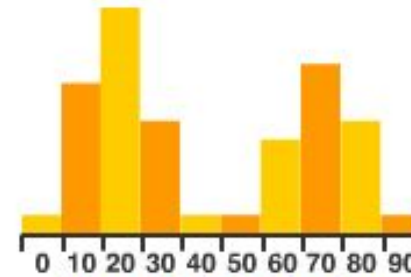
**skewed distribution**  
negatively skewed,  
skewed left



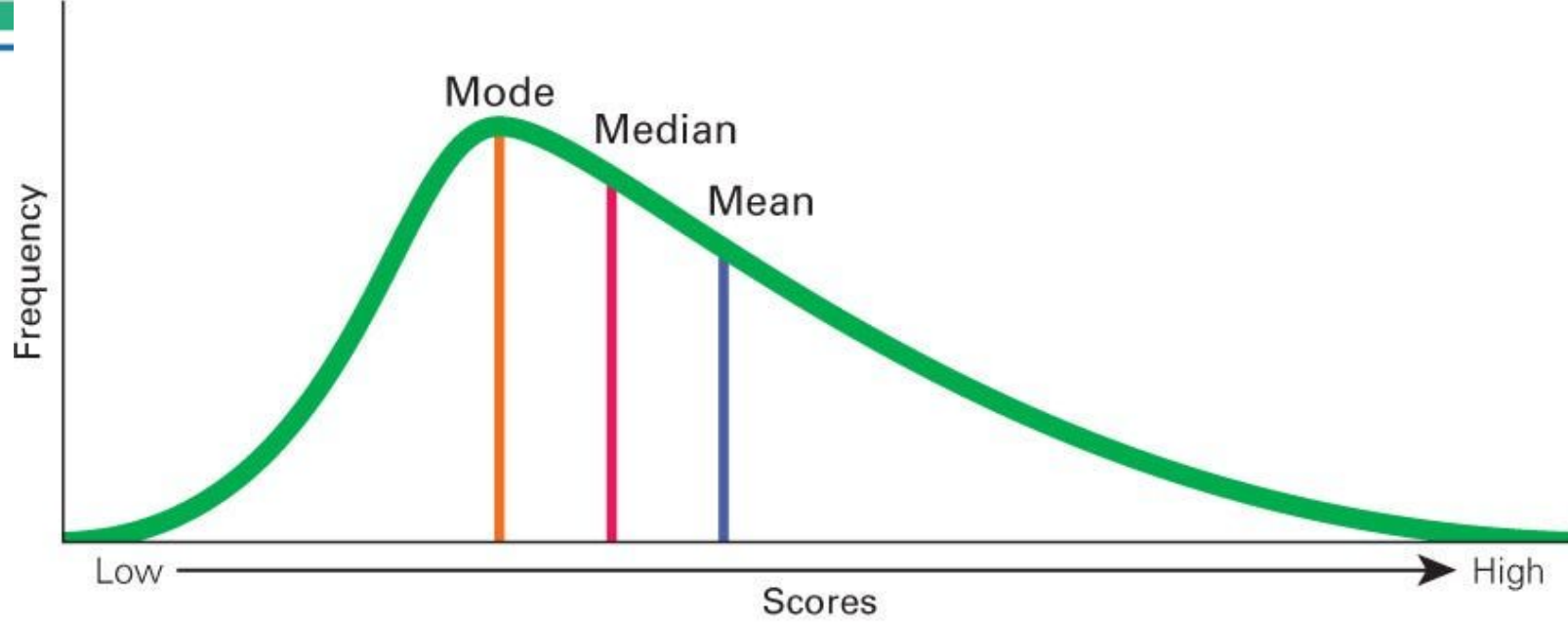
**uniform distribution**  
equally spread,  
no peaks



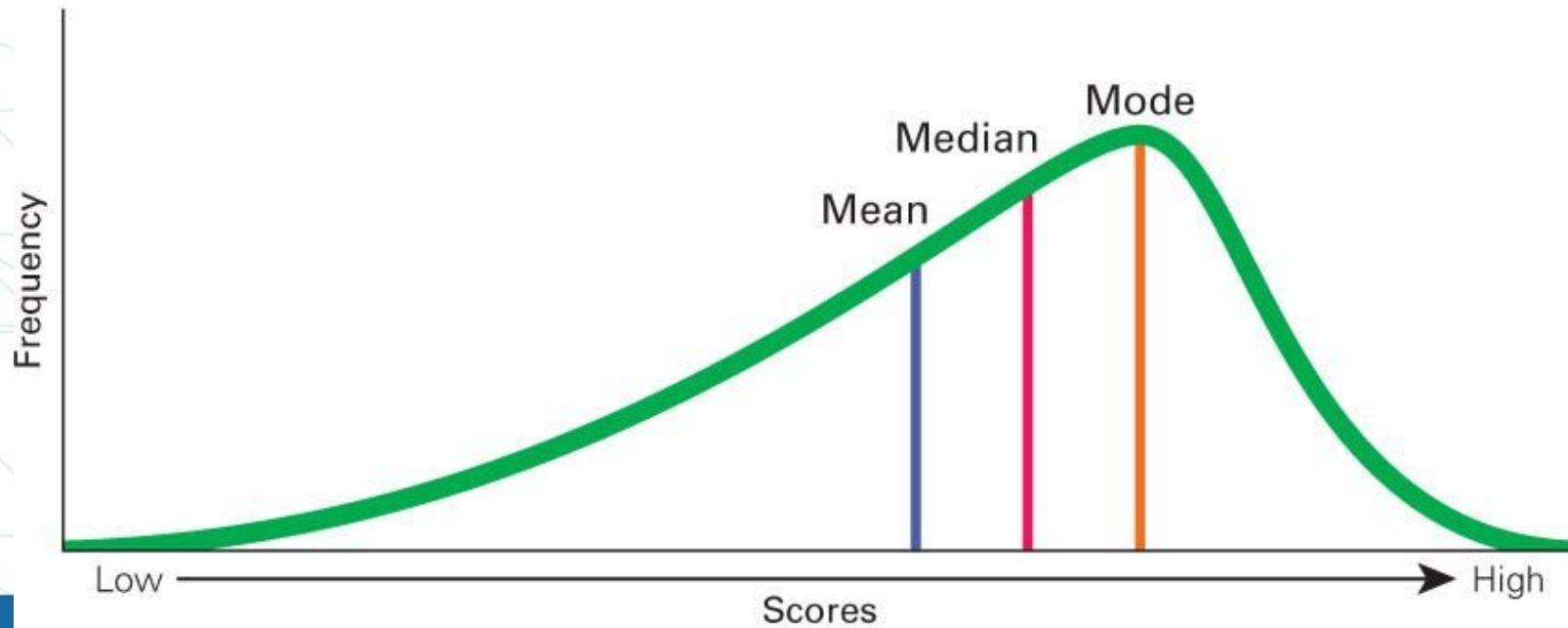
**bimodal distribution**  
two modes,  
symmetric



**bimodal distribution**  
two modes,  
non-symmetric



(a) Right-skewed distribution



(b) Left-skewed distribution

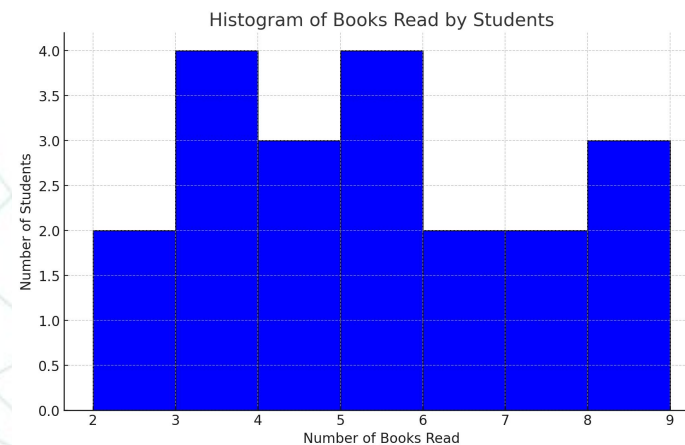
## Data Distribution and Visualization

### Histograms

A histogram is a graphical representation that organizes a group of data points into user-specified ranges. It shows the frequency distribution of a dataset.

### Example

Create a histogram for a dataset representing the number of books read by students in a class.



- Histograms help visualize the shape of the data distribution (e.g., normal, skewed) and identify patterns like bimodality or skewness.

# Probability Basics (1 hour)

## Probability Basics

fundamental probability concepts, including the rules of probability, conditional probability, and Bayes' theorem. These concepts are widely used in AI and machine learning, particularly in classification problems, predictive modeling, and decision-making processes.





## Basic Probability Concepts

### Definition of Probability

Probability measures the likelihood of an event occurring. It ranges from 0 to 1, where 0 indicates an impossible event and 1 indicates a certain event.

### Example

The probability of getting heads in a fair coin toss is 0.5.

### Random Events and Outcomes

#### Random Event

- An event whose occurrence cannot be predicted with certainty. Examples include rolling a die, flipping a coin, or picking a card from a deck.

#### Outcomes

- The possible results of a random event. For a coin toss, the possible outcomes are heads or tails. For rolling a six-sided die, the outcomes are 1, 2, 3, 4, 5, and 6.



## Basic Probability Concepts

### Probability Formula

For an event A, the probability  $P(A)$  is calculated as:

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

### Example

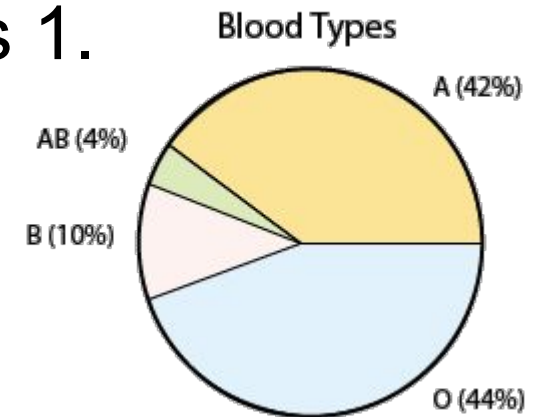
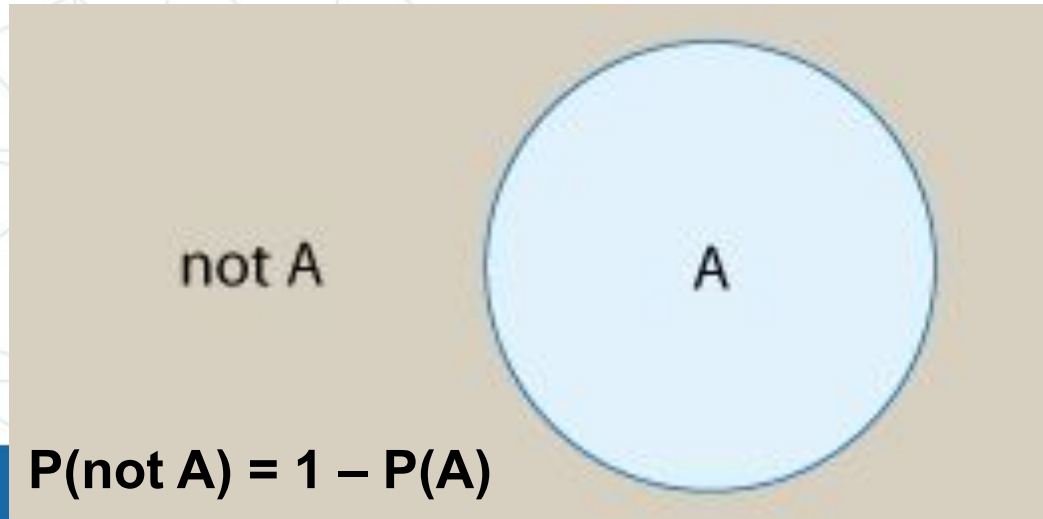
The probability of drawing a heart from a standard deck of cards:

$$P(\text{heart}) = \frac{13}{52} = \frac{1}{4} = 0.25$$

# Elementary Probability

A probability gives the likelihood that a defined event will occur. It is quantified as a positive number between 0 (the event is impossible) and 1 (the event is certain). Thus, the higher the probability of a given event, the more likely it is to occur.

- For any event A,  $0 \leq P(A) \leq 1$ .
- The sum of the probabilities of all possible outcomes is 1.
- 



## Probability Rules

### Addition Rule (For Mutually Exclusive Events)

If two events A and B cannot happen at the same time (i.e., they are mutually exclusive), the probability that either A or B will occur is:

$$P(A \cup B) = P(A) + P(B)$$

### Example

### Example of the Addition Rule for Mutually Exclusive Events:

Let's take the example of rolling a six-sided die.

- Event A is rolling a 2.
- Event B is rolling a 5.

Since these two events cannot happen at the same time (you can't roll both a 2 and a 5 on a single roll of the die), they are mutually exclusive events.

According to the addition rule for mutually exclusive events, the probability of rolling either a 2 or a 5 is the sum of the probabilities of each event:

$$P(A \text{ or } B) = P(A) + P(B)$$

The probability of rolling a 2 is:

$$P(A) = \frac{1}{6}$$

The probability of rolling a 5 is:

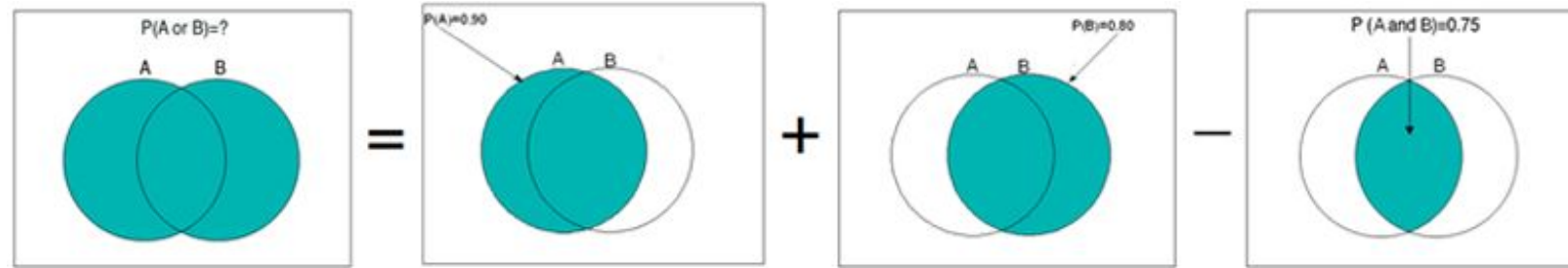
$$P(B) = \frac{1}{6}$$

So, the probability of rolling either a 2 or a 5 is:

$$P(A \text{ or } B) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

Thus, the probability of rolling a 2 or a 5 on a six-sided die is  $\frac{1}{3}$  or approximately 0.333.

## Addition Rule



We therefore get:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

*addition rules (union of events)*

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

*mutually exclusive*  $P(A \cup B) = P(A) + P(B)$



## Probability Rules

### Multiplication Rule (For Independent Events)

If two events A and B are independent (the occurrence of one does not affect the occurrence of the other), the probability that both A and B will occur is:

$$P(A \cap B) = P(A) \times P(B)$$

### Example

The probability of getting heads when flipping a coin and rolling a 4 on a six-sided die:

$$P(\text{heads and } 4) = P(\text{heads}) \times P(4) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$$



*multiplication rules (joint probability)*

*dependent*  $P(A \cap B) = P(A) * P(B|A)$

*independent*  $P(A \cap B) = P(A) * P(B)$

*mutually exclusive*  $P(A \cap B) = 0$

$P(B|A)$  “the probability of B happening *given* A has occurred

## Conditional Probability Formula

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Probability of  $A$  given  $B$

Probability of  $A$  and  $B$

Probability of  $B$

A math teacher gave her class two tests. 80% of the class passed the first test. 60% of the class passed both tests. What percent of those who passed the first test also passed the second test?

$$P(\text{second given first}) = P(\text{first and second}) / P(\text{first}) = 0.6/0.8 = 75\%$$

## Conditional Probability Formula

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## Bayes' Theorem

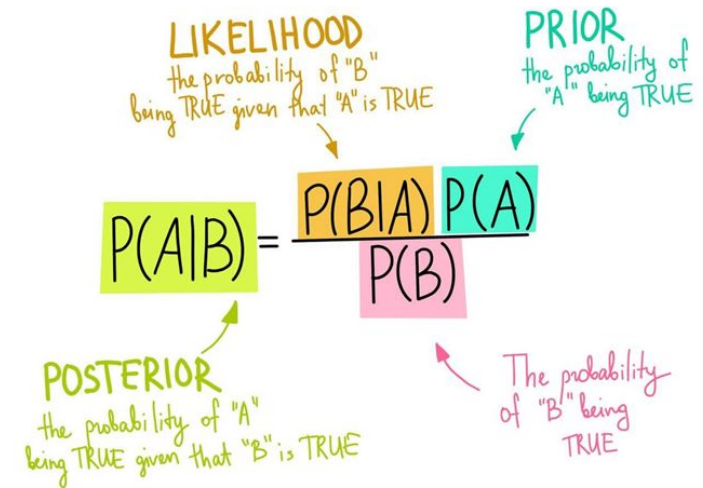
**Introduction to Bayes' Theorem:** Bayes' theorem allows us to update our probability estimates based on new **evidence** or **information**. It is particularly useful in AI for making predictions based on **prior knowledge** and new data.

### Bayes' Theorem Formula:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Where:

- $P(A|B)$  is the posterior probability: the probability of event  $A$  given  $B$  has occurred.
- $P(B|A)$  is the likelihood: the probability of event  $B$  given  $A$  is true.
- $P(A)$  is the prior probability: the probability of event  $A$  before observing  $B$ .
- $P(B)$  is the marginal likelihood: the total probability of event  $B$ .





## Bayes' Theorem

### Example

#### Given Information:

- The probability that a person has the disease is 1% of the population:

$$P(\text{Disease}) = 0.01$$

- The test correctly identifies the disease 99% of the time (test sensitivity):

$$P(\text{Positive Test} \mid \text{Disease}) = 0.99$$

- The test produces false positives 5% of the time (false positive rate):

$$P(\text{Positive Test} \mid \text{No Disease}) = 0.05$$

- The probability that a person does not have the disease is 99%:

$$P(\text{No Disease}) = 0.99$$

#### Problem:

We want to calculate the probability that a person has the disease given that they tested positive, i.e.,  $P(\text{Disease} \mid \text{Positive Test})$ .

## Bayes' Theorem

### Example

Using Bayes' Theorem:

Bayes' Theorem gives us the following formula:

$$P(\text{Disease} \mid \text{Positive Test}) = \frac{P(\text{Positive Test} \mid \text{Disease}) \times P(\text{Disease})}{P(\text{Positive Test})}$$

Calculating  $P(\text{Positive Test})$ :

$$P(\text{Positive Test}) = P(\text{Positive Test} \mid \text{Disease}) \times P(\text{Disease}) + P(\text{Positive Test} \mid \text{No Disease}) \times P(\text{No Disease})$$

Substituting the values:

$$P(\text{Positive Test}) = (0.99 \times 0.01) + (0.05 \times 0.99) = 0.0099 + 0.0495 = 0.0594$$

Apply Bayes' Theorem:

$$P(\text{Disease} \mid \text{Positive Test}) = \frac{0.99 \times 0.01}{0.0594} \approx 0.1667$$

This means there is approximately a 16.67% chance that the person actually has the disease, despite testing positive. This highlights how Bayes' theorem can provide more accurate probability estimates when additional information is available.



# Introduction to Inferential Statistics (30 mins)

## Introduction to Inferential Statistics

### What is Inferential Statistics?

Inferential statistics involve using data from a sample to make generalizations about a larger population. **Unlike** descriptive statistics, which summarize data, inferential statistics allow us to draw conclusions and make predictions.

### Purpose in AI and Data Science

- Helps estimate population parameters (e.g., the mean, proportion).
- Allows for hypothesis testing to determine if observed patterns are likely to be genuine or if they occurred by chance.
- Facilitates predictions and decision-making based on sample data.

## Key Concepts of Inferential Statistics

### Sampling Distributions

A sampling distribution is the probability distribution of a given statistic (e.g., mean, proportion) based on a random sample.

### Importance

- Sampling distributions help understand how the sample statistic (like the sample mean) varies from sample to sample.
- They form the basis for making inferences about the population.

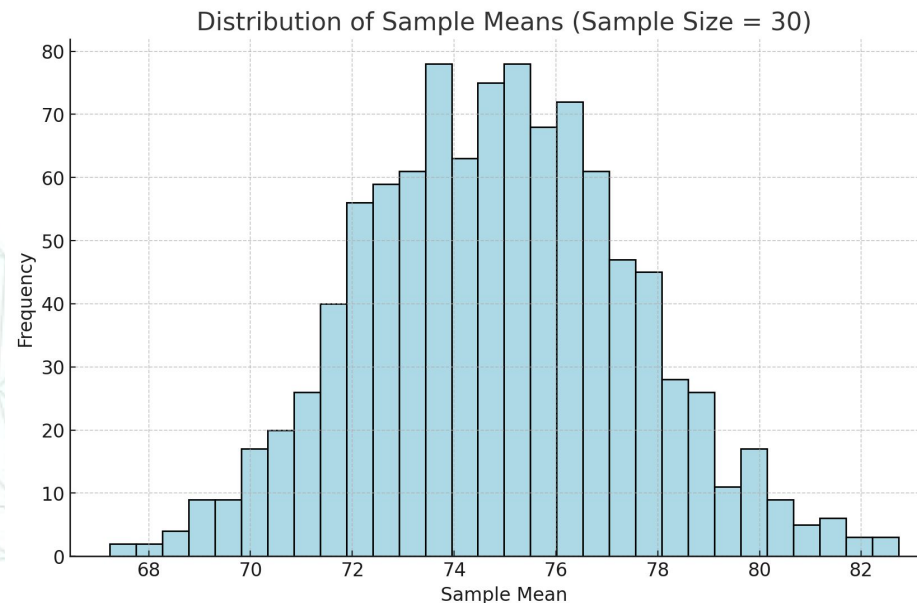
### Example

If we take multiple random samples of students' test scores and calculate the mean for each sample, the distribution of those sample means would form a sampling distribution. This distribution tends to be normal, especially as the sample size increases, according to the Central Limit Theorem.

## Key Concepts of Inferential Statistics

### Example

If we take multiple random samples of students' test scores and calculate the mean for each sample, the distribution of those sample means would form a sampling distribution. This distribution tends to be normal, especially as the sample size increases, according to the Central Limit Theorem.



## Key Concepts of Inferential Statistics

### Central Limit Theorem (CLT)

The CLT states that the sampling distribution of the sample mean will be approximately normally distributed, regardless of the population's distribution, provided the sample size is sufficiently large (usually  $n > 30$ ).

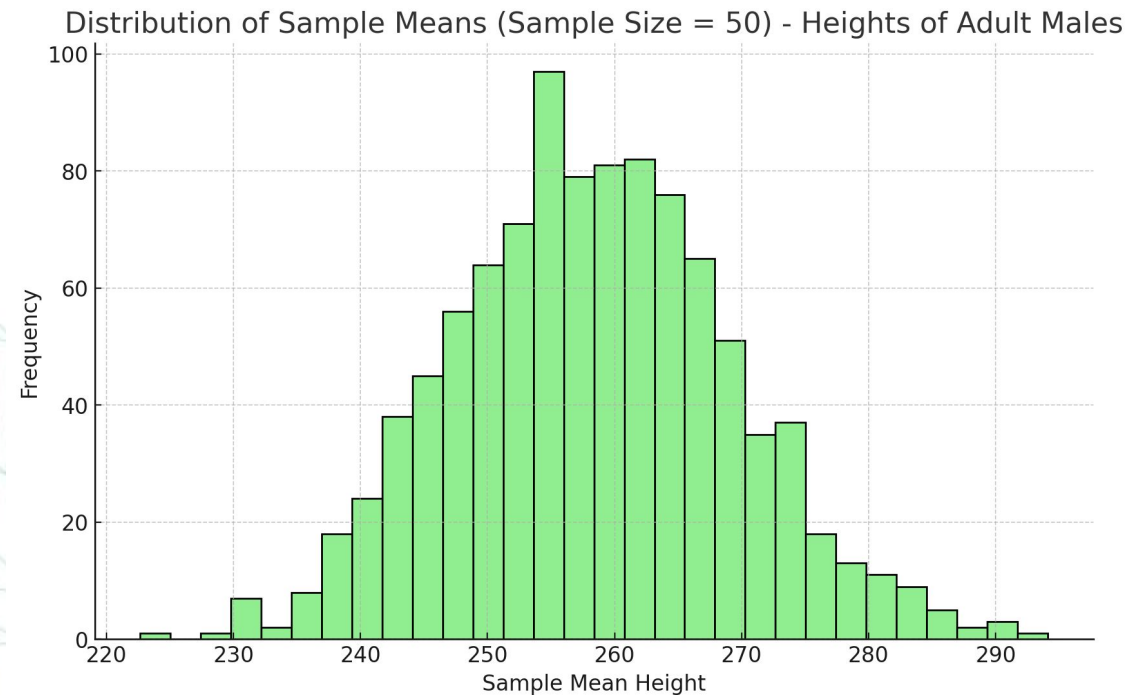
### Significance

The CLT enables us to use the normal distribution to make inferences about population parameters even when the population distribution is not normal.

## Key Concepts of Inferential Statistics

### Example

Suppose we are analyzing the average height of adult males. Even if the population distribution of heights is skewed, the mean of sufficiently large samples will follow a normal distribution.





## Hypothesis Testing

Hypothesis testing is a statistical method used to make inferences about population parameters based on sample data. It helps us decide whether there is enough evidence to reject a null hypothesis.

### Key Terms

- **Null Hypothesis ( $H_0$ ):** A statement that there is no effect or no difference. It serves as the default assumption. For example, "The mean score of students is equal to 75."
- **Alternative Hypothesis ( $H_1$  or  $H_a$ ):** The statement that contradicts the null hypothesis. For example, "The mean score of students is not equal to 75."
- **P-value:** The probability of observing the sample data, or something more extreme, assuming the null hypothesis is true. A low p-value (typically  $< 0.05$ ) suggests that the observed data is unlikely under the null hypothesis, leading to its rejection.
- **Significance Level ( $\alpha$ ):** A threshold chosen before conducting the test, often set at 0.05, indicating a 5% risk of concluding that a difference exists when there is no actual difference.

## Hypothesis Testing

### Steps in Hypothesis Testing

1. **State the Hypotheses:** Define the null and alternative hypotheses.
2. **Select a Significance Level ( $\alpha$ ):** Commonly set at 0.05.
3. **Collect Data and Calculate a Test Statistic:** Based on the sample data, calculate a statistic (e.g., t-statistic, z-statistic) that measures the difference.
4. **Find the P-value:** Compare the test statistic to the theoretical distribution to find the p-value.
5. **Make a Decision:**
  - If the p-value  $< \alpha$ , reject the null hypothesis (evidence suggests an effect).
  - If the p-value  $\geq \alpha$ , do not reject the null hypothesis (insufficient evidence to suggest an effect).

## Hypothesis Testing

### Example

A company claims that their employees work an average of 8 hours per day. A random sample of 30 employees shows an average of 7.5 hours with a standard deviation of 1 hour. We could conduct a hypothesis test to determine if this sample provides enough evidence to reject the company's claim.

## Confidence Intervals

A confidence interval provides a range of values that is likely to contain the population parameter (e.g., mean) with a certain level of confidence (e.g., 95%).

**Formula :**  $\text{Confidence Interval} = \bar{x} \pm Z \left( \frac{\sigma}{\sqrt{n}} \right)$

- $\bar{x}$  = Sample mean
- $Z$  = Z-value (from standard normal distribution for a given confidence level)
- $\sigma$  = Population standard deviation (or sample standard deviation if population standard deviation is unknown)
- $n$  = Sample size

### Interpretation

A 95% confidence interval means that if we took 100 different samples and computed a confidence interval for each sample, approximately 95 of the intervals would contain the true population mean.

## Confidence Intervals

### Example

If a sample mean score on a test is 70 with a standard deviation of 5 and sample size of 30, a 95% confidence interval for the mean score might be calculated. This interval provides a range within which we expect the true mean score for the entire population to fall.

# Math Basics

Session 2: (Linear Algebra  
Fundamentals)



## Agenda:

|   |                                        |
|---|----------------------------------------|
| 1 | Vectors and Matrices                   |
| 2 | Matrix Operations                      |
| 3 | Linear Transformations and Eigenvalues |

# Vectors and Matrices

## Vector

A vector is an ordered list of numbers (scalars) that can represent a point in space. Vectors have both **magnitude** (length) and **direction**. In the context of AI, vectors are used to **represent data points, features, or weights**.

Vectors are typically represented as **columns** (column vectors) or **rows** (row vectors). For example, a 3-dimensional vector can be written as:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

## Vectors

### Example

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- **Position Vector:** In a 2D space, a vector  $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$  can represent a point 3 units to the right and 4 units up from the origin.
- **Feature Vector:** In machine learning, a vector can represent features of a data point, such as  $\mathbf{x} = \begin{bmatrix} \text{age} \\ \text{income} \end{bmatrix} = \begin{bmatrix} 30 \\ 50000 \end{bmatrix}$ .

## Operations on Vectors

### Addition

Vectors of the same dimension can be added by adding corresponding elements:

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}$$

## Operations on Vectors

### Subtraction

Similar to addition, but subtracting corresponding elements.

### Scalar Multiplication

Multiplying a vector by a scalar (number) scales each element of the vector:

$$c \cdot \mathbf{v} = c \cdot \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} c \cdot v_1 \\ c \cdot v_2 \end{bmatrix}$$



## Matrices

A matrix is a rectangular array of numbers arranged in rows and columns. Matrices can represent multiple vectors, data tables, or linear transformations.

A matrix  $A$  with  $m$  rows and  $n$  columns is denoted as  $A_{m \times n}$ . For example, a 2x3 matrix looks like this:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

## Matrices

### Examples

- **Data Matrix:** In a dataset, each row could represent a different data point (case), and each column could represent a different feature (variable).
- **Transformation Matrix:** Matrices can be used to perform transformations like rotations, scaling, or translations in space

## Why Vectors and Matrices are Important in AI and Machine Learning ?

- **Data Representation:** Vectors and matrices are used to represent datasets, where each row is a data point and each column is a feature.
- **Linear Transformations:** Matrices can represent transformations applied to data, such as scaling, rotating, and translating data points in space.
- **Matrix Operations:** Essential for performing calculations in machine learning algorithms, such as linear regression, neural networks, and principal component analysis (PCA).

## Summary

- Vectors represent points or directions in space and can be used to model features of data points.
- Matrices represent more complex data structures and transformations and are fundamental tools for organizing and manipulating data in AI.
- Understanding vector and matrix operations is crucial for implementing and optimizing machine learning algorithms.

# Matrix Operations

## Matrix Operations

This segment covers essential matrix operations that are fundamental for understanding how data is manipulated and transformed in AI and machine learning. Matrix operations enable the efficient computation of complex transformations, solving systems of linear equations, and performing various data manipulations.



## Matrix Multiplication

Matrix multiplication is a way of combining two matrices to produce a new matrix. The number of columns in the first matrix must match the number of rows in the second matrix for multiplication to be possible.

### Notation:

If  $A$  is an  $m \times n$  matrix and  $B$  is an  $n \times p$  matrix, the product  $AB$  is an  $m \times p$  matrix.

### Formula:

Each element of the resulting matrix  $C=AB$  is calculated as the dot product of a row in  $A$  and a column in  $B$ :

$$c_{ij} = \sum_{k=1}^n a_{ik} \cdot b_{kj}$$

## Matrix Multiplication Example

- e • Let  $A$  be a  $2 \times 3$  matrix and  $B$  be a  $3 \times 2$  matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$$

- The resulting matrix  $C = AB$  is:

$$C = \begin{bmatrix} (1 \cdot 7 + 2 \cdot 9 + 3 \cdot 11) & (1 \cdot 8 + 2 \cdot 10 + 3 \cdot 12) \\ (4 \cdot 7 + 5 \cdot 9 + 6 \cdot 11) & (4 \cdot 8 + 5 \cdot 10 + 6 \cdot 12) \end{bmatrix} = \begin{bmatrix} 58 & 64 \\ 139 & 154 \end{bmatrix}$$

## Determinant of a Matrix

The determinant is a scalar value that can be computed from the elements of a square matrix. It provides important properties of the matrix, such as whether it is invertible (non-singular).

### Notation:

For a square matrix  $A$ , the determinant is denoted as  $\det(A)$  or  $|A|$ .

### Formula for a 2x2 Matrix

$$\det \left( \begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = ad - bc$$

## Determinant of a Matrix

### Exempl

For a matrix  $A = \begin{bmatrix} 3 & 8 \\ 4 & 6 \end{bmatrix}$ , the determinant is:

$$\det(A) = (3 \cdot 6) - (8 \cdot 4) = 18 - 32 = -14$$

## Inverse of a Matrix

The inverse of a matrix  $A$  is a matrix  $A^{-1}$  such that when it is multiplied by  $A$ , it results in the identity  $I$ :  $A \times A^{-1} = I$

### Conditions

- A matrix must be square (same number of rows and columns) and have a non-zero determinant to have an inverse.

### Formula for a 2x2 Matrix

For a matrix  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ , the inverse is:

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

## Inverse of a Matrix

**Example**

For  $A = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$ :

$$\det(A) = (4 \cdot 6) - (7 \cdot 2) = 24 - 14 = 10$$

$$A^{-1} = \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}$$



## Transpose of a Matrix

The transpose of a matrix is obtained by flipping the matrix over its diagonal, converting rows into columns.

### Notation:

If  $A$  is a matrix, the transpose is denoted by  $A^T$

### Example

For  $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ :

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

## Summary

- **Matrix multiplication** is crucial for data transformations and neural network operations.
- **Determinants** help determine the properties of matrices, such as invertibility.
- **Matrix inverses** are used to solve linear equations and optimize models.
- **Matrix transposition** is essential for aligning data dimensions and preparing matrices for multiplication.

## Types of Matrices

### Square Matrix

A matrix with the same number of rows and columns (e.g.,  $n \times n$ ). It's used often in linear transformations and finding determinants.

Example: 
$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

## Types of Matrices

### Diagonal Matrix

A square matrix where all non-diagonal elements are zero. Only the diagonal elements may be non-zero.

Example: 
$$\begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$$

## Types of Matrices

### Identity Matrix

A special type of diagonal matrix where all the diagonal elements are 1. It acts as the multiplicative identity in matrix operations.

Example:  $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

## Types of Matrices

### Zero Matrix

A matrix where all elements are zero. It's the additive identity in matrix operations.

Example:

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$



## Types of Matrices

### Row Matrix

A matrix with only one row.

Example:  $[1 \ 2 \ 3]$

## Types of Matrices

### Symmetric Matrix

A matrix with only one row.

A square matrix that is equal to its  
trans

Example:

$$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$$

$$(A = A^T)$$

# Linear Transformations and Eigenvalues

## Linear Transformations and Eigenvalues

This segment covers the concepts of linear transformations and eigenvalues, which are fundamental in understanding how data can be manipulated and analyzed in various dimensions. These concepts play a crucial role in areas such as dimensionality reduction, stability analysis, and optimization in AI and machine learning.

## Linear Transformations

A linear transformation is a function that maps vectors from one space to another, preserving vector addition and scalar multiplication. Essentially, it transforms vectors while maintaining the structure of the space.

### Matrix Representation

Matrix Representation Linear transformations can be represented by matrices. If  $x$  is a vector and  $A$  is a matrix, then  $Ax$  is a linear transformation of  $x$ .

**Example** : If  $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$  and  $x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ , then:

$$Ax = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$$

## Common Types of Linear Transformations

### Scaling

Changes the size of vectors. Represented by a diagonal matrix where each diagonal element scales the corresponding dimension.

**Example** Scaling by a factor of 2,  $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ , scales vectors by 2.  
:



## Common Types of Linear Transformations

### Rotation

Rotates vectors around the origin. Represented by a rotation matrix.

**Example in 2D:**  $A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$  rotates vectors by an angle  $\theta$ .

## Common Types of Linear Transformations

### Reflection

Flips vectors over a specified line (in 2D) or plane (in 3D).

**Example in 2D:** Reflection over the x-axis,  $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ .

## Common Types of Linear Transformations

### Shearing

Distorts the shape of vectors in one direction while keeping the other direction unchanged.

**Example in 2D** Horizontal shear in 2D,  $A = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$ , where  $k$  is the shear factor.

## Eigenvalues and Eigenvectors

In the context of a linear transformation, an eigenvector of a matrix  $A$  is a non-zero vector that only changes by a scalar factor when  $A$  is applied to it. The scalar is known as the eigenvalue.

**Eigenvalue ( $\lambda$ ):** A scalar indicating how much the eigenvector is stretched or compressed.

**Eigenvector ( $v$ ):** A vector that does not change its direction during the transformation.

# Eigenvalues and Eigenvectors

## Mathematical Formulation

$$A\mathbf{v} = \lambda\mathbf{v}$$

Where:

- **A** is the transformation matrix.
- **v** is the eigenvector.
- **λ** is the eigenvalue.

## Eigenvalues and Eigenvectors

### Finding Eigenvalues and Eigenvectors

- To find eigenvalues, solve the characteristic equation:

$$\det(A - \lambda I) = 0$$

- Once eigenvalues ( $\lambda$ ) are known, solve  $(A - \lambda I)v = 0$  to find the corresponding eigenvectors.



## Eigenvalues and Eigenvectors

**Example:** Let  $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ :

1. Calculate the characteristic equation:

$$\det(A - \lambda I) = \det \begin{bmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{bmatrix} = (4 - \lambda)(3 - \lambda) - 2 \cdot 1 = 0$$

2. Solve for  $\lambda$ :

$$\lambda^2 - 7\lambda + 10 = 0 \Rightarrow (\lambda - 5)(\lambda - 2) = 0$$

So, the eigenvalues are  $\lambda=5$  and  $\lambda=2$ .

3. Substitute each  $\lambda$  back into  $(A - \lambda I)v = 0$  to find the eigenvectors.